

ANDREA VALLA

Senior Full-Stack Software Engineer / Software Architect | TypeScript, React, Node.js, PostgreSQL & AI Systems

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Professional Summary

Senior Full-Stack Software Engineer and Software Architect with 20+ years of experience designing, building, and operating production software. Hands-on work across TypeScript, React, Node.js/Bun, PostgreSQL, GraphQL, distributed systems, API integrations, CI/CD, and cloud infrastructure. Recent projects include AI/LLM systems, agent orchestration, ecommerce platforms, data pipelines, and Manufacturing Execution Systems (MES).

Technical Skills

- Programming Languages:** TypeScript, JavaScript, C#, PHP, SQL
- Frontend:** React, React Native, Apollo, URQL, Redux, Vite, Tailwind CSS
- Backend & APIs:** Node.js, Bun, Deno, Express, .NET, GraphQL, REST APIs
- Databases & Data:** PostgreSQL, Supabase, Redis, MongoDB, SQL Server, MySQL, BigQuery, Airbyte, Cube.js
- Architecture:** Software Architecture, System Design, Distributed Systems, API Integration, Third-Party Integrations, Background Jobs, Event-Driven Systems
- DevOps & Cloud:** Linux/Unix, Docker, GitHub Actions, AWS, GCP, Azure, Cloudflare, CI/CD
- Testing:** Automated Testing, Vitest, pgTAP, Database & RLS Testing
- AI / LLM:** LLMs, AI Agents, Agent Orchestration, OpenAI API, LangChain, Model Routing, LLM Gateways, Usage/Cost Metering, Tool Permissions
- Industrial:** Manufacturing Execution Systems (MES), IIoT, OPC UA, MQTT, ERP Integration, ETL

Professional Experience

Solution Architect, Full-Stack Developer & System Administrator

BrandsDistribution.com, Turin, Italy Feb 2022 - Sep 2026

- Built queue-backed BDroppy integrations with Shopify, Squarespace, Wix, and EKM supporting 50,000+ connected stores; shipped apps through the Shopify and Wix app marketplaces.
- Designed and built AutoEpoque, a classic-car marketplace for classifieds and auctions using React, TypeScript, Supabase/PostgreSQL, Row-Level Security (RLS), BullMQ/Redis background jobs, and pgTAP database tests.
- Architected and built Aidify, an AI-powered chatbot platform integrated with ecommerce channels.
- Built a BI and analytics platform using BigQuery, Airbyte, Cube.js, and Looker Studio to support cost and ROI analysis.
- Owned software architecture, backend, frontend, integrations, data, and production operations across multiple products.

TypeScript, React, Node.js, Bun, PostgreSQL, Supabase, Redis, BigQuery, GraphQL

CTO & Co-founder

Agile Factory, Turin, Italy **Mar 2018 - Jan 2022**

- Co-founded and led engineering of a real-time Manufacturing Execution System (MES) for SMEs, from product strategy and software architecture through implementation, customer deployment, and production operations.
- Designed Industrial Internet of Things (IIoT) and ERP integration architecture using OPC UA, MQTT, and ETL pipelines to connect industrial equipment with customer business systems.
- Built two React applications and Node.js/GraphQL services using Redis, Docker, automated testing, and code-quality tooling.

Node.js, React, GraphQL, Redis, Docker, OPC UA, MQTT

Full-Stack Developer

Industrial Cloud, Politecnico di Torino spin-off, Turin, Italy **Jun 2015 - Mar 2018**

- Delivered manufacturing web applications from analysis and product requirements through implementation, customer deployment, and ongoing updates.

Node.js, React, Elasticsearch, MySQL, PHP

Full-Stack Developer

Intesa Sanpaolo, Moncalieri, Italy **Sep 2010 - Jun 2015**

- Built business applications, ETL/data workflows, and reporting systems using SQL Server, SSIS/SSRS, .NET, and JavaScript.

Developer & System Administrator

Reply S.p.A., Turin, Italy **Sep 2003 - Sep 2010**

- Developed software and administered Linux, Unix, and Microsoft systems, including constrained enterprise infrastructure and production-risk containment.

Unix System Administrator

Global Value Services, Turin, Italy **Jul 2002 - Jul 2003**

- Managed systems and applications supporting FCA web properties in HP-UX environments.

Open Source & Selected Projects

AI Office

Creator **2026 - Present**

- Built an open-source multi-agent orchestration platform for AI agents, workflows, tasks, pipelines, and durable executions.
- Implemented concurrency control, task locking, LLM/model routing, audit logs, tool permissions, and governance.
- Integrated OpenAI, Codex, and Claude Code behind provider-neutral interfaces.
- Designed pluggable persistence and knowledge layers using PostgreSQL, SQLite, and SurrealDB, with extensible agents and domain behavior.

TypeScript, Bun, PostgreSQL, SQLite, SurrealDB, OpenAI, Codex, Claude Code

AutoEpoque

Product architecture & engineering at BrandsDistribution **2025 - Present**

- Own end-to-end architecture and implementation for a classic-car marketplace supporting classifieds and auctions.
- Built a React/TypeScript monorepo on Supabase/PostgreSQL with Row-Level Security (RLS), BullMQ/Redis jobs, media workflows, internationalization, and pgTAP database and security tests.

TypeScript, React, PostgreSQL, Supabase, Redis, BullMQ, pgTAP

Spoken Languages

Italian Native; **English** Fluent; **German** Basic